

	University of Central Punjab Faculty of Information Technology & Computer Sciences Data Structures, Spring 2026 Quiz 3 V1	
	Total Marks: 20	Time Allowed: 30 min
Section	Student Name	Student Registration
D5		

CLO #	CLO Statement	Taxonomy Level	PLO
2	Implement various data structures and apply them in simple applications.	C2	4

Question # 1: A doubly linked list is maintained using only a head pointer. Write a function that moves the node present at a given position $P = 3$ to the end of the doubly linked list while maintaining the relative order of all remaining nodes.

Note: You are NOT allowed to create a new node or copy data from one node to another. Only **one traversal** is allowed and no tail pointer maintained.

Sample Output:

Original List: 10 < - > 20 < - > 30 < - > 40 < - > 50

Position = 3

Updated List: 10 < - > 20 < - > 40 < - > 50 < - > 30

Question # 2: Write a recursive program to determine whether a sentence is a palindrome while ignoring spaces and case sensitivity.